Workshop – Week 1 – Some selected solutions

1. In each case, **given a truth table**, find an equivalent Boolean expression, then draw an associated logic circuit.

NOTE: it is a good idea to use an online logic circuit simulator such as https://logic.ly/demo/

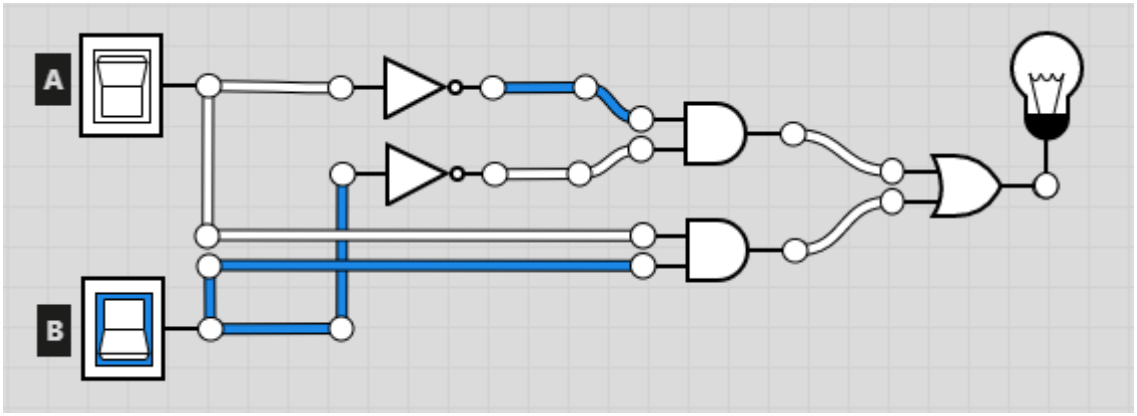
Solutions:

a)

A	B	output
F	F	T
F	T	F
T	F	F
T	T	T

~A~B+AB

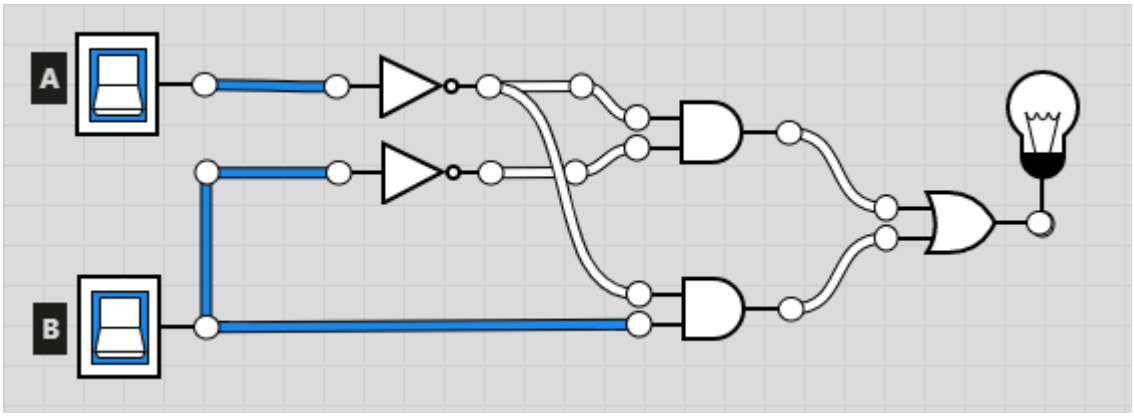
b) ~A~B+~AB+AB = ~A(~B+B)+AB
= ~A+AB = ~A+B



c)

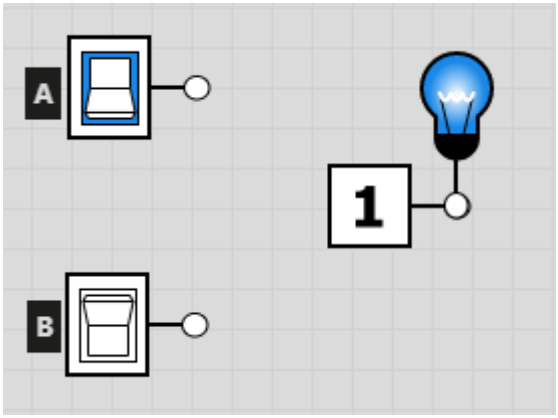
A	B	output
0	0	1
0	1	1
1	0	0
1	1	0

~A~B+~AB = ~A(~B+B) = ~A



d) What would you do if the outputs of the truth table were all 1 (or all T)?

What if the outputs of the truth table were all 0 (or all F)?



When outputs are all 1, the Boolean expression is 1, using 1 constant in circuit.

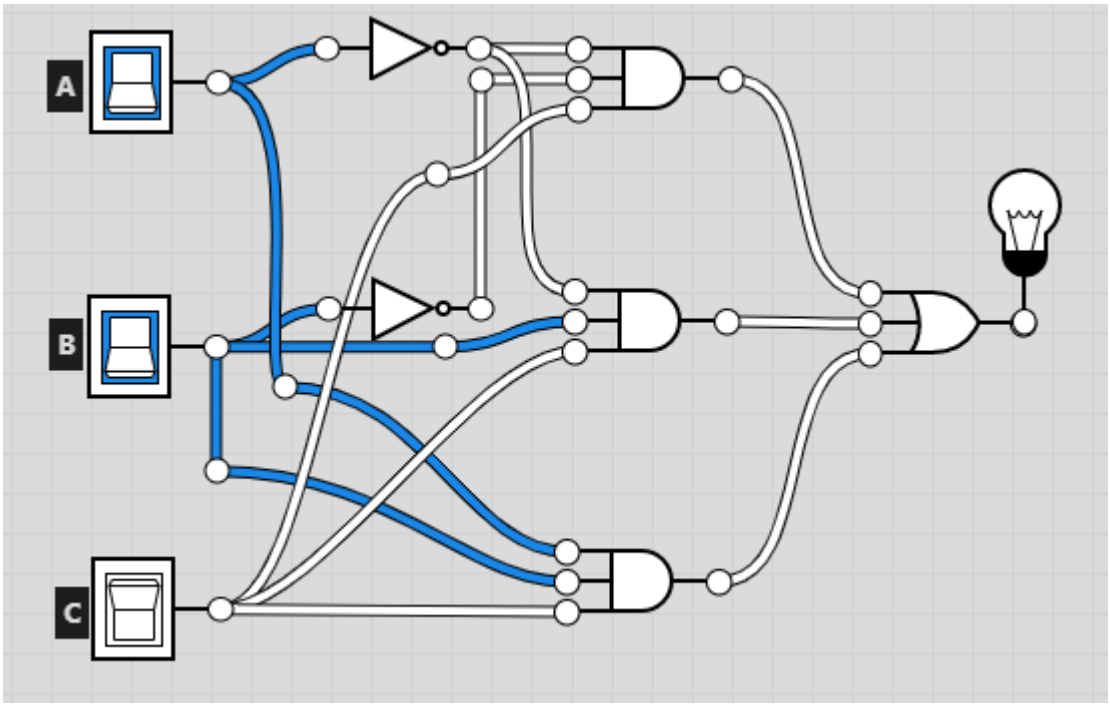
~A~B+~AB+A~B+AB = ~A(~B+B)+A(~B+B)
= ~A+A = 1

When outputs are all 0, the Boolean expression is 0, using 0 constant in circuit.

e)

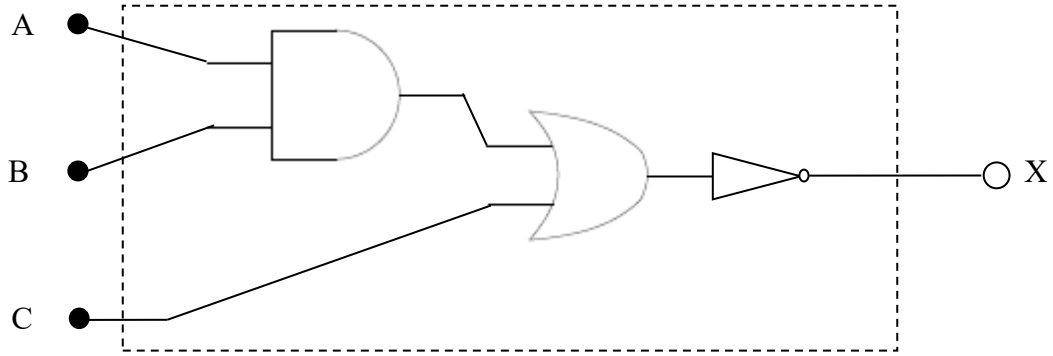
A	B	C	X
F	F	F	F
F	F	T	T
F	T	F	F
F	T	T	T
T	F	F	F
T	F	T	F
T	T	F	F
T	T	T	T

~A~BC+~ABC+ABC = ~AC(~B+B)+ABC = ~AC+ABC = C(~A+AB) = C(~A+B) = ~AC+BC



f) ~A~BC+~AB~C+~ABC+A~BC+AB~C = ~AC(~B+B)+B~C(~A+A)+A~BC = ~AC+B~C+A~BC = C(~A+A~B)+B~C = C(~A+~B)+B~C = ~AC+~BC+B~C

2. Given the logic circuit below, find the associated truth table and Boolean expression:



Working...

A	B	C	A.B	(A.B) + C	X
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	0	1
0	1	1	0	1	0
1	0	0	0	0	1
1	0	1	0	1	0
1	1	0	1	1	0
1	1	1	1	1	0

The final truth table (you could use T and F if you prefer)**:**

A	B	C	X
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

~A~B~C+~AB~C+A~B~C = ~B~C(~A+A)+~AB~C = ~B~C+~AB~C = ~C(~B+~AB) = ~C(~B+~A) = ~C ~(AB) = ~(AB+C)

And the Boolean expression from this truth table is:

$X = \sim A \bullet \sim B \bullet \sim C + \sim A \bullet B \bullet \sim C + A \bullet \sim B \bullet \sim C$

But actually the circuit provides us with a simpler (and equivalent) Boolean expression:

$X = \sim((A \bullet B) + C)$

3. Given the following Boolean expression, draw the circuit and set up the truth table:

X = ~A ● B ● C + ~A ● B ● ~C + A ● ~B ● C

Working...

Make the truth table by using the usual repeating pattern for A, B and C. Then for X (the output) place a 1 (or T) next to every row that matches one of the terms in the Boolean expression.

The final truth table (you could use T and F if you prefer)**:**

A	B	C	X
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	0

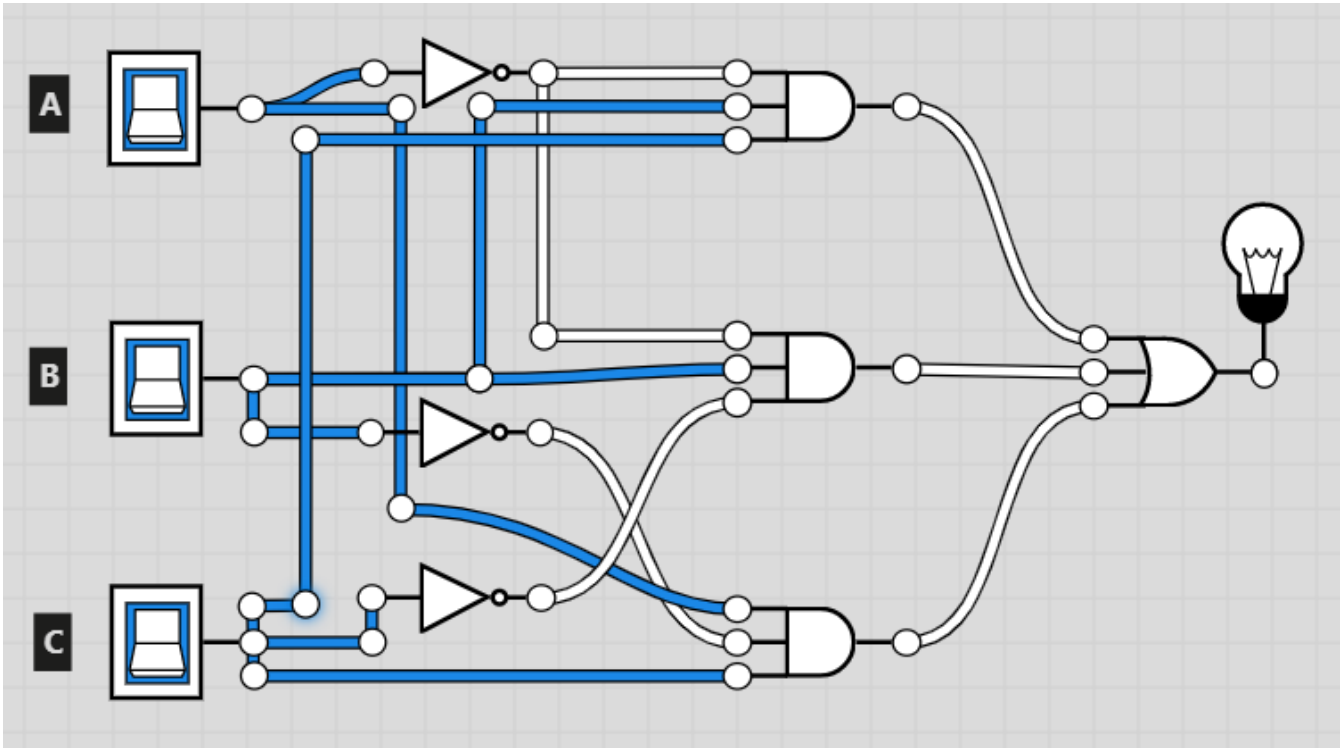
this row matches the second term

this row matches the first term

this row matches the third term

~ABC+~AB~C+A~BC =
~AB(C+~C)+A~BC =
~AB+A~BC

Use the Boolean expression (or the truth table) to set up the circuit:



4. a) A=1, B=0, C=0 => D=1
4. b) A=1, B=1, C=1 => D=1